

SGD, Momentum, and Adam

Deep Learning

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Contents of This Video

In this video, we will cover:

- Stochastic Gradient Descent (SGD) and mini-batches
- Momentum: accelerating convergence
- Adam: adaptive learning rates
- Comparing optimizer behaviors
- Practical guidelines for choosing optimizers

Stochastic Gradient Descent (SGD)

Full-batch gradient descent:

$$w \leftarrow w - \alpha \frac{1}{n} \sum_{i=1}^n \nabla L_i$$

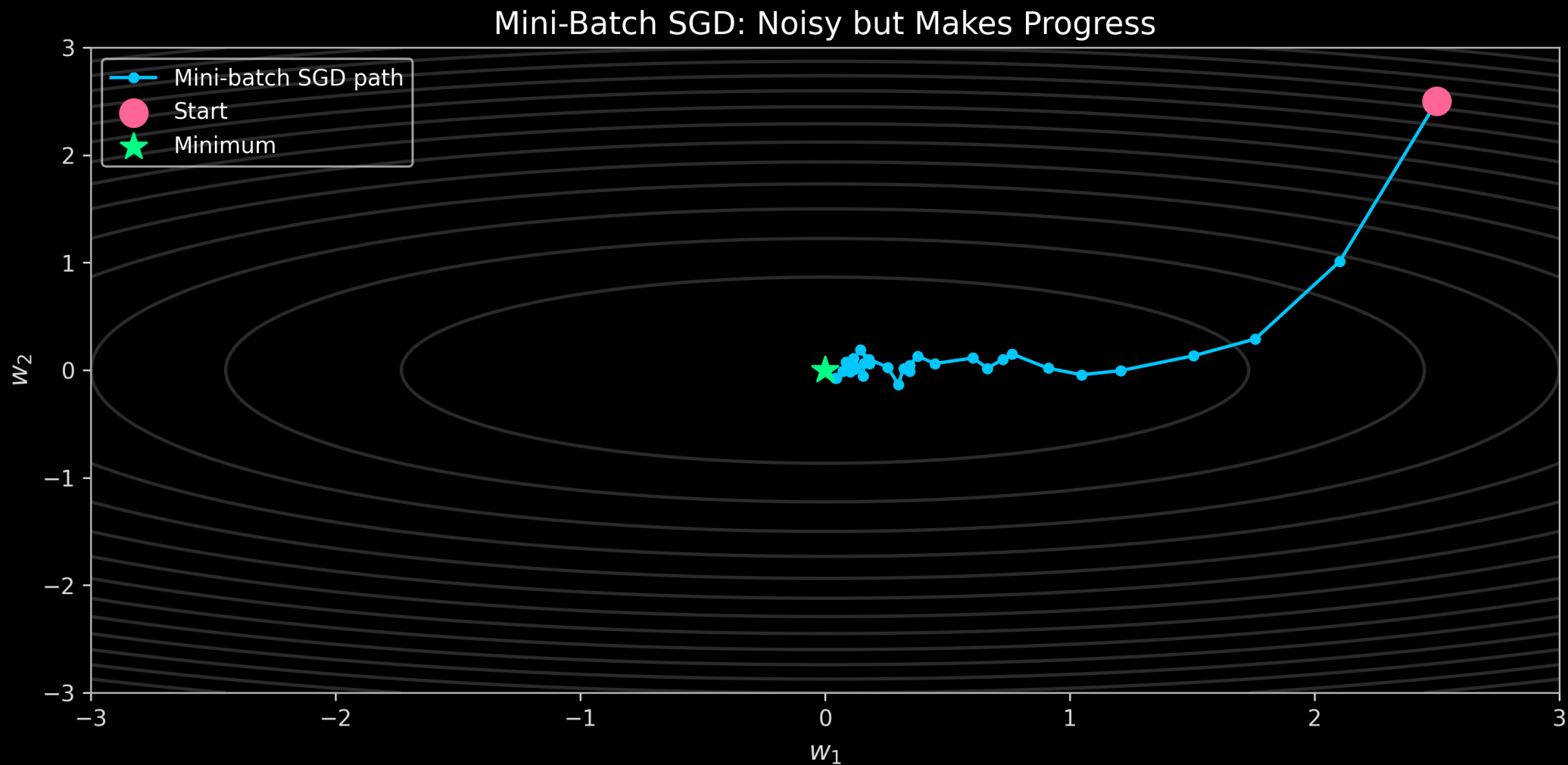
Stochastic gradient descent:

$$w \leftarrow w - \alpha \nabla L_i \quad (\text{one random sample})$$

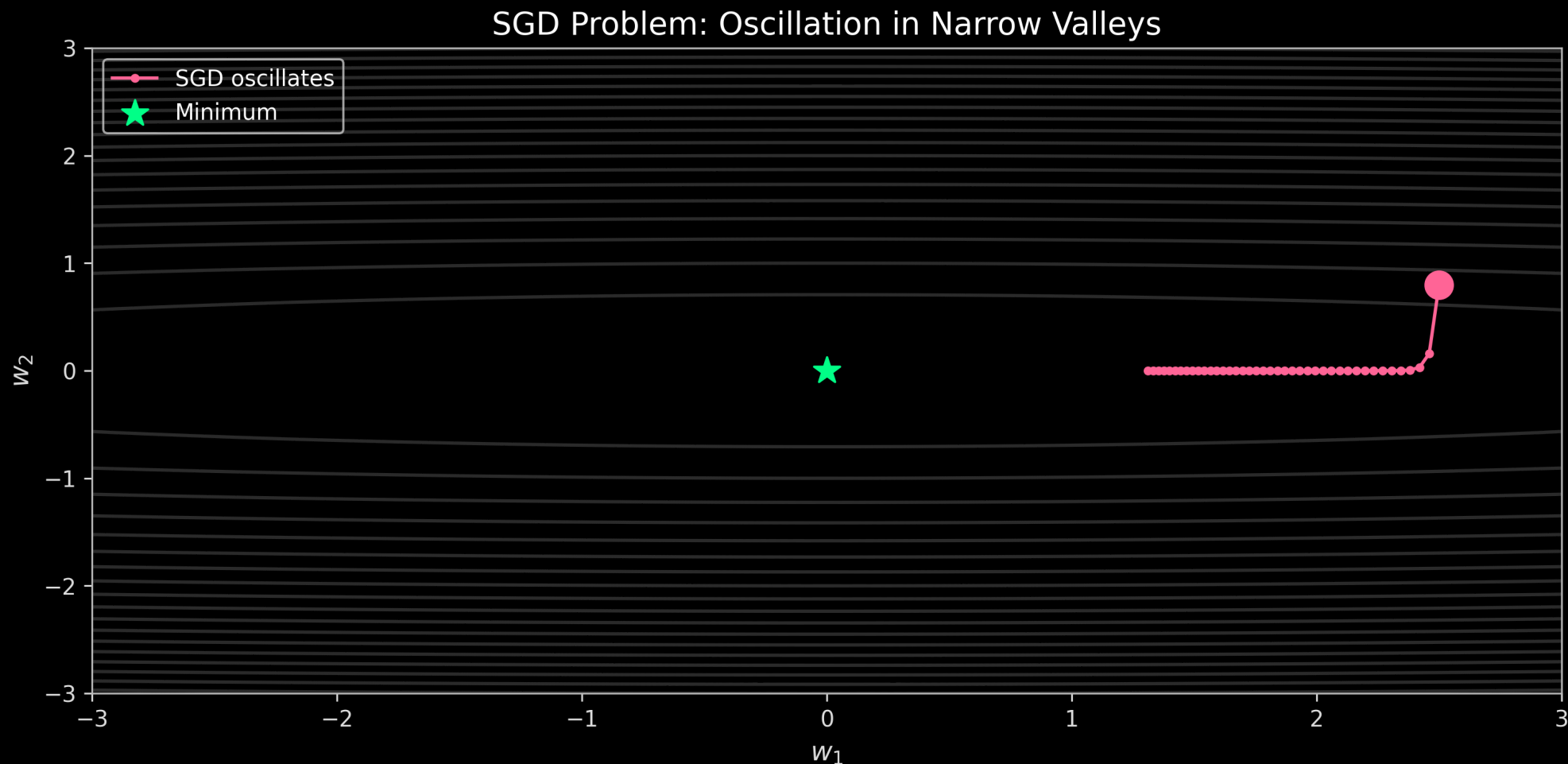
Mini-batch gradient descent (most common):

$$w \leftarrow w - \alpha \frac{1}{B} \sum_{i \in \text{batch}} \nabla L_i$$

Mini-Batch SGD in Practice



The Problem with SGD



Gradients are steep in w_2 direction, shallow in w_1

SGD oscillates across the valley instead of going straight to minimum

Momentum: Building Up Speed

Add velocity that accumulates gradient direction:

$$v_t = \beta v_{t-1} + \nabla L(w_t)$$

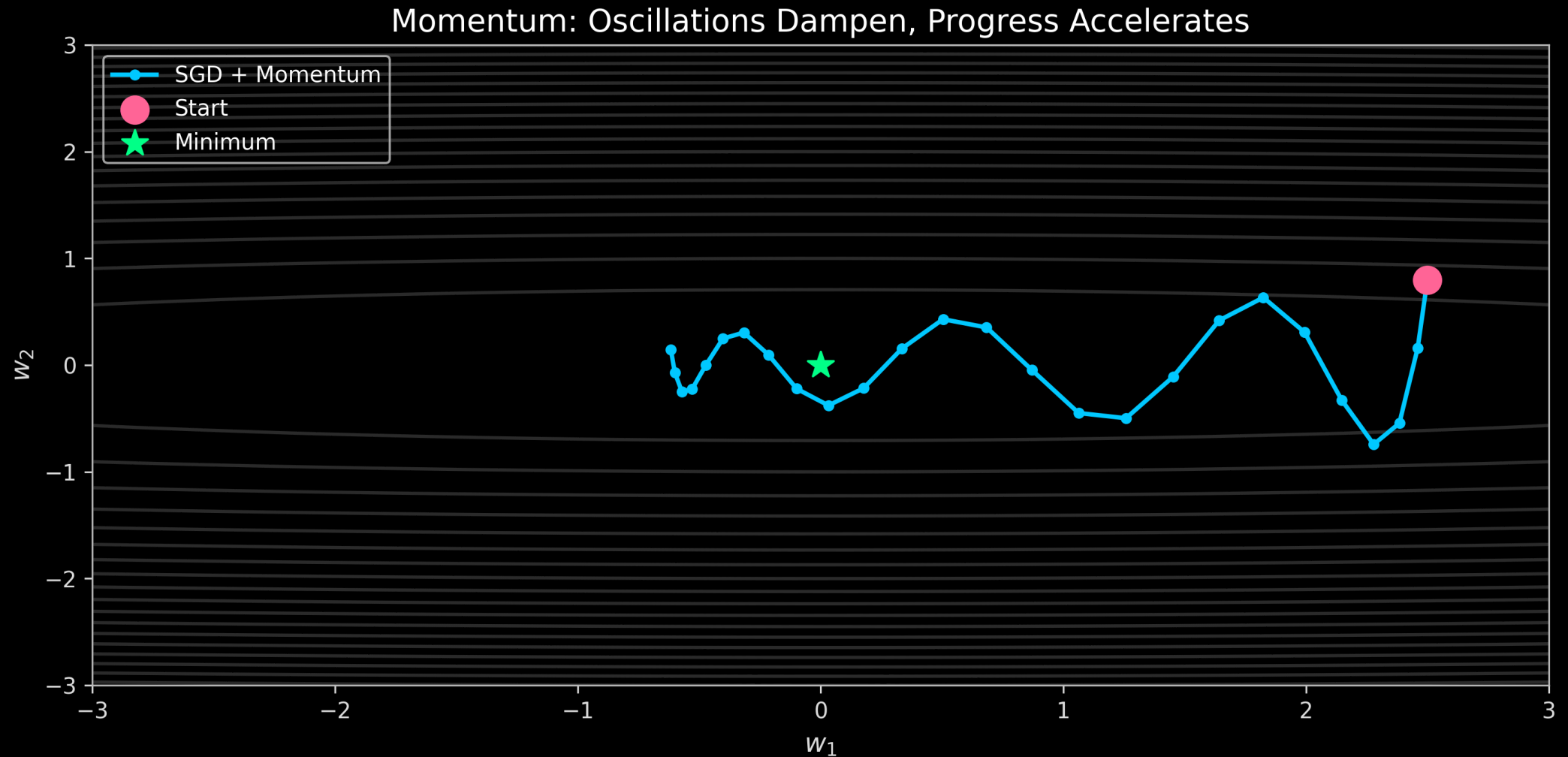
$$w_{t+1} = w_t - \alpha v_t$$

Where:

- $\beta \approx 0.9$ is the momentum coefficient
- v is the velocity (accumulated gradient)

Intuition: Ball rolling downhill gains speed in consistent direction

Momentum in Action



Adam: Adaptive Learning Rates

Key idea: Different parameters need different learning rates

Adam (Adaptive Moment Estimation) maintains:

- First moment (mean of gradients): $m_t = \beta_1 m_{t-1} + (1 - \beta_1)g_t$
- Second moment (variance of gradients): $v_t = \beta_2 v_{t-1} + (1 - \beta_2)g_t^2$

Update:

$$w_{t+1} = w_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

Adam: The Default Choice

Hyperparameters:

- $\alpha = 0.001$ (learning rate)
- $\beta_1 = 0.9$ (first moment decay)
- $\beta_2 = 0.999$ (second moment decay)
- $\epsilon = 10^{-8}$ (numerical stability)

Why Adam is popular:

- Works well with default hyperparameters
- Adapts to different parameter scales
- Fast convergence on most problems
- Robust to noisy gradients

Comparing Optimizers

Optimizer	Pros	Cons
SGD	Simple, can generalize well	Slow, sensitive to LR
SGD + Momentum	Faster, dampens oscillations	Extra hyperparameter
Adam	Fast, adaptive, robust	May generalize worse than SGD

Practical guidance:

- Start with Adam ($lr=0.001$)
- If overfitting, try SGD + momentum with tuning
- Research shows: SGD sometimes finds better minima

What We've Covered

In this video, we've learned:

- Mini-batch SGD: balance between speed and accuracy
- Momentum: accumulate gradients to dampen oscillations
- Adam: adaptive learning rates per parameter
- Adam is the default choice ($\text{lr}=0.001$, $\beta_1=0.9$, $\beta_2=0.999$)
- SGD + momentum may generalize better with careful tuning